torch.block_diag#

https://pytorch.org/docs/stable/generated/torch.block_diag.html#torch.block_

Description:

Create a block diagonal matrix from provided tensors. *tensors – One or more tensors with 0, 1, or 2 dimensions. A 2 dimensional tensor with all the input tensors arranged in order such that their upper left and lower right corners are diagonally adjacent. All other elements are set to 0.

Function Signature

```
torch.block_diag(*tensors)[source]#
Parameters
Returns
Return type
```

Returns:

Tensor

Code Examples

```
>>> import torch
>>> A = torch.tensor([[0, 1], [1, 0]])
>>> B = torch.tensor([[3, 4, 5], [6, 7, 8]])
>>> C = torch.tensor(7)
>>> D = torch.tensor([1, 2, 3])
>>> E = torch.tensor([[4], [5], [6]])
>>> torch.block_diag(A, B, C, D, E)
tensor([[0, 1, 0, 0, 0, 0, 0, 0, 0, 0],
        [1, 0, 0, 0, 0, 0, 0, 0, 0, 0],
        [0, 0, 3, 4, 5, 0, 0, 0, 0, 0],
        [0, 0, 6, 7, 8, 0, 0, 0, 0, 0],
        [0, 0, 0, 0, 0, 7, 0, 0, 0, 0],
        [0, 0, 0, 0, 0, 0, 1, 2, 3, 0],
        [0, 0, 0, 0, 0, 0, 0, 0, 0, 4],
        [0, 0, 0, 0, 0, 0, 0, 0, 0, 5],
        [0, 0, 0, 0, 0, 0, 0, 0, 0, 6]])
```

Detailed Documentation

Documentation

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